

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0279

Gvrd Sewage 100 kV
July 03, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0279. The Interconnection Customer proposes a 127.5 MW solar photovoltaic facility interconnecting at the Gvrd Sewage 100 kV substation in the Illinois area. The request was received on October 21, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 168 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$8,793,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Ann to Gvrd Sewage 100 KV: 126.2% loading (107 MVA) following the outage of the Deltrail to Fielder Creek 100 KV L; DFAX = 72.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ann to Gvrd Sewage 100 KV L	Rebuild line with higher-capacity conductor	\$11,180,000	79% (\$8,793,000)
	Total allocated to project		\$8,793,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.